

AI-Powered UTI Antibiotic System

ANALYSIS REPORT • ACADEMIC RESEARCH DOCUMENT

Patient Demographics

Age / Gender	40.0 / Female
Department	Gynaecology
Diagnosis	Urethritis
UTI Classification	Recurrent UTI
Sample Type	Pus Swab

Clinical Presentation

Chief Complaints: Burning urination;Pelvic discomfort

Comorbidities: None

Surgical History: None

Social History: Non smoker

Laboratory Results

Test Name	Value	Unit	Normal Range
Cbp Lymphocytes	38.0	%	20 - 40
Wbc	8950.0	cells/ μ L	4000 - 11000
Polymorphs	63.0	%	40 - 75
Crp	8.0	mg/L	< 10
Rft Serum Creatinine	0.7	mg/dL	0.6 - 1.3
Serum Uric Acid	4.0	mg/dL	3.5 - 7.2
Blood Urea	24.0	mg/dL	7 - 20
Cue Pus Cells	11.0	/HPF	0 - 5
Epithelial Cells	3.0	/HPF	0 - 5
Proteins	Trace		Negative
Rbc	1.0	/HPF	0 - 3

AI Pathogen Prediction

Predicted Organism	Enterococcus faecium
Bacteria Type	Gram-positive coccus (Higher risk of Vancomycin resistance - VRE)

Antibiotic Susceptibility Profile

Predicted Sensitive	Daptomycin
Predicted Resistant	Clarithromycin

Recommended Therapy

Daptomycin

Dosage: 6 mg/kg IV once daily (e.g., 400 mg IV daily for a 70-kg patient)

Precautions: Monitor serum CPK weekly for myopathy; adjust dose in severe renal impairment (eGFR <30 mL/min, give 4 mg/kg). Ensure no known hypersensitivity to daptomycin. No dose adjustment needed for current creatinine 0.7 mg/dL. Avoid concomitant use with statins unless monitored.

Rationale: Enterococcus faecium is predicted to be vancomycin-resistant (VRE). Daptomycin is the only antibiotic in the predicted sensitive list and retains activity against VRE, making it the appropriate choice for this recurrent urethritis.

Antibiotic Drug History

Clinical Summary

Infection Summary:

A 40-year-old female presents with recurrent urethritis, confirmed by pus swab culture. The causative organism is predicted to be *Enterococcus faecium*, a Gram-positive coccus with a higher risk of vancomycin resistance (VRE).

Resistance and Sensitivity Profile:

The organism is predicted to be resistant to Clarithromycin and sensitive to Daptomycin.

Recommended Antibiotic:

Daptomycin is prescribed at 6 mg/kg IV once daily (e.g., 400 mg IV daily for a 70-kg patient). This is the only antibiotic in the sensitive list and is effective against VRE, making it the appropriate choice for this recurrent infection.

Precautions and Considerations:

- Monitor serum CPK weekly to detect myopathy.
- Adjust dose in severe renal impairment (eGFR <30 mL/min; reduce to 4 mg/kg).
- Ensure no hypersensitivity to daptomycin.
- Avoid concomitant use with statins unless closely monitored.
- No dose adjustment is needed for the current serum creatinine of 0.7 mg/dL.

This regimen addresses the infection while minimizing risks associated with resistance and adverse effects.